

A universal counterexample to the renorming converse for asymptotic centers

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Source conjecture

The source is C. Angosto, M. C. Listán-García, and F. Rambla-Barreno, *Continuity properties of sequentially asymptotically center-complete spaces* [1], available as arXiv:1403.4646 and later published in RACSAM.

For a bounded sequence $\bar{x} = (x_n)$ in a real Banach space, it defines

$$\text{ar}(\bar{x}) = \inf_{z \in X} \limsup_n \|x_n - z\|, \quad \text{ac}(\bar{x}) = \left\{ z : \limsup_n \|x_n - z\| = \text{ar}(\bar{x}) \right\}.$$

Writing $C_n(\bar{x}) = \{x_m : m \geq n\}$, its tail pseudometric is

$$d(\bar{x}, \bar{y}) = \inf\{\varepsilon > 0 : \text{for every } n \text{ there is } m \text{ such that } C_m(\bar{x}) \subset C_n(\bar{y}) + \varepsilon B_X, \quad C_m(\bar{y}) \subset C_n(\bar{x}) + \varepsilon B_X\}.$$

Conjecture 3.3 asks whether equality of both asymptotic centers and radii under *every equivalent renorming* forces $d(\bar{x}, \bar{y}) = 0$. The answer is no in every nonzero Banach space.

Combining [8] and [9] we get that $d(\bar{x}_i, \bar{y}) < \frac{1}{2^{i-1}}$. ■

In the proposition and conjecture that follows we try to advocate that this distance is, in certain sense, “sharp” concerning centers.

Proposition 3.2 *Let $\bar{x}, \bar{y}$ be bounded sequences in a Banach space X satisfying $d(\bar{x}, \bar{y}) = 0$. Then*

1. $\text{ac}(\bar{x}) = \text{ac}(\bar{y})$ and $\text{ar}(\bar{x}) = \text{ar}(\bar{y})$.
2. $d(\bar{x}, \bar{y}) = 0$ for each equivalent renorming of X .

PROOF

Take $z \in X$. For every $\varepsilon > 0$ we have $d(\bar{x}, \bar{y}) < \varepsilon$ and this implies $|\overline{\lim}_n \|x_n - z\| - \overline{\lim}_n \|y_n - z\|| < \varepsilon$. Since ε is arbitrary we deduce that

$$\overline{\lim}_n \|x_n - z\| = \overline{\lim}_n \|y_n - z\|$$

which yields immediately $\text{ar}(\bar{x}) = \text{ar}(\bar{y})$ and $\text{ac}(\bar{x}) = \text{ac}(\bar{y})$.

The second statement is a direct consequence of the definition of d . ■

Is there a sort of converse to the previous proposition?

Conjecture 3.3 *Let $\bar{x}, \bar{y}$ be bounded sequences in a Banach space X . If $\text{ac}(\bar{x}) = \text{ac}(\bar{y})$ and $\text{ar}(\bar{x}) = \text{ar}(\bar{y})$ hold for each equivalent renorming of X , then $d(\bar{x}, \bar{y}) = 0$.*

Universal counterexample

Theorem 1. *Let X be any nonzero real Banach space. There are bounded sequences $\bar{x}, \bar{y}$ such that, for every equivalent norm q on X ,*

$$\limsup_n q(x_n - z) = \limsup_n q(y_n - z) \quad (z \in X), \quad (1)$$

but

$$d(\bar{x}, \bar{y}) > 0.$$

In particular, the sequences have the same asymptotic center and radius for every equivalent renorming, so Conjecture 3.3 is false. In fact, equality (1) is stronger than the conjecture's hypothesis.

Proof. Fix $u \in X \setminus \{0\}$ and take the periodic sequences

$$\bar{x} = (-u, u, -u, u, \dots), \quad \bar{y} = (-u, 0, u, -u, 0, u, \dots).$$

Let q be any norm on X ; equivalence to the original norm is not needed for this part. For every $z \in X$,

$$\begin{aligned} R_x(z) &:= \limsup_n q(x_n - z) = \max\{q(z - u), q(z + u)\}, \\ R_y(z) &:= \limsup_n q(y_n - z) = \max\{q(z - u), q(z), q(z + u)\}. \end{aligned}$$

Since $2z = (z - u) + (z + u)$, the triangle inequality gives

$$q(z) \leq \frac{q(z - u) + q(z + u)}{2} \leq \max\{q(z - u), q(z + u)\}.$$

Thus $R_x = R_y$ pointwise, proving (1). Their common radius is $q(u)$: the triangle inequality applied to $2u = (z + u) - (z - u)$ gives $R_x(z) \geq q(u)$, while equality holds at $z = 0$. Their common center is therefore

$$\{z \in X : q(z - u) \leq q(u), q(z + u) \leq q(u)\}.$$

Every tail of $\bar{x}$ has value set $A = \{-u, u\}$, and every tail of $\bar{y}$ has value set $B = \{-u, 0, u\}$. Hence the definition of d , computed in the original norm, reduces to the Hausdorff distance between A and B . Since $A \subset B$ and $\text{dist}(0, A) = \|u\|$,

$$d(\bar{x}, \bar{y}) = \|u\| > 0.$$

This contradicts the conclusion of Conjecture 3.3. □

The one-dimensional instance $X = \mathbb{R}$, $u = 1$ already suffices: the two periodic value sets are $\{-1, 1\}$ and $\{-1, 0, 1\}$ and their distance is one. The all-space form shows that the failure is not caused by a shortage of renormings in dimension one.

The structural obstruction: convex-hull blindness

The example is an instance of a general obstruction.

Lemma 1. *Let A be a finite subset of a normed space. For every $z \in X$,*

$$\sup_{a \in \text{conv } A} \|a - z\| = \max_{a \in A} \|a - z\|.$$

Consequently, periodic sequences enumerating finite sets A and B with $\text{conv } A = \text{conv } B$ have identical full asymptotic-radius functions under every norm, although their tail pseudodistance is the Hausdorff distance between A and B and can be positive.

Proof. If $v = \sum_j \lambda_j a_j$ is a convex combination of points in A , then

$$\|v - z\| \leq \sum_j \lambda_j \|a_j - z\| \leq \max_{a \in A} \|a - z\|.$$

The reverse inequality is immediate from $A \subset \text{conv } A$. For a periodic sequence, each tail contains its complete finite value set, so the remaining claims follow directly. $\square$

Here $A = \{-u, u\}$ and $B = A \cup \{0\}$ have the same convex hull. Thus no amount of equivalent renorming can make center/radius data recover the nonconvex tail geometry measured by d . Even replacing the conjecture’s center-and-radius hypothesis with equality of the entire functions $z \mapsto \limsup_n \|x_n - z\|$ for every equivalent norm would not repair it.

Verification, novelty, and scope

The proof is exact and uses only periodic sequences, the triangle inequality, and the definition of d . The review-critical identities are independently checkable from the two finite tail sets. No compactness, center-existence theorem, or numerical claim is used.

On 2026-08-12, exact-title, exact-conjecture, and exact-renorming-phrase searches found the arXiv and published source and later work citing the paper for center-continuity results, but no resolution or correction of Conjecture 3.3. The four cheap run indexes contained no result for arXiv:1403.4646. This is a bounded novelty check, not an assertion that every non-arXiv source has been exhausted.

The result fully refutes Conjecture 3.3 and explains the obstruction through convex hulls. It does not address the paper’s separate question whether every codimension-one subspace of a $C(K)$ space is sequentially asymptotically center-complete.

Human review recommendation

Review as a likely-valid full counterexample. The decisive observation is pointwise, not merely minimax: adding the midpoint 0 to the periodic tail set does not change its farthest-distance function in any norm, while it changes the tail Hausdorff geometry by exactly $\|u\|$.

References

- [1] C. Angosto, M. C. Listán-García, and F. Rambla-Barreno, *Continuity properties of sequentially asymptotically center-complete spaces*, arXiv:1403.4646; RACSAM **110** (2016), 809–822.